

Project Assignment

Due Date: May 8th, 2025 (11:59pm).

Implementation, Analysis, and Final Reporting

Assignment Tasks:

1. Model Implementation and Data Preparation:

- **Dataset:** Work on publicly available dataset or your dataset if you have one. Choose a dataset so that the running time of your model is reasonable (e.g., few hours at most).
- **Develop and Train Models:** Implement proposed deep learning techniques/models on a dataset.
- **Data Processing:** Prepare and preprocess your dataset(s), including augmentation, normalization, and handling of missing values or noise. Justify the choices made during preprocessing.
- **Reproducibility:** Ensure that your implementation is reproducible by documenting all settings (e.g., hyperparameters, seeds for randomness) and dependencies (e.g., libraries, hardware specifications).

2. Evaluation and Results Analysis:

- **Performance Metrics:** Evaluate the model's performance using appropriate statistical and domain-specific metrics. For example, accuracy, F1-score, precision, recall, or RMSE as applicable. Fine-tune your models and combine them into a great solution.
- **Comparison:** Compare your results with baseline methods or state-of-the-art models. Explore many different models and short-list the best ones. Fine-tune the hyperparameters using grid search.
- **Visualization:** Provide clear visualizations (e.g., charts, confusion matrices, loss curves) to support your analysis.

Submission Instructions:

1. Deliverables:

- Final implementation code and documentation.
- Presentation summarizing key findings (3-5 slides).

2. Submission: Upload all deliverables to Blackboard.

Evaluation Criteria:

- Maximum number of students per group is four.
- **Implementation Quality (35%):** Correctness, reproducibility, and innovation in implementation.
- **Results Analysis (30%):** Depth of evaluation and clarity in presenting findings.
- **Presentation and Questions (25%):** Ability to communicate key insights, contributions, and answer questions.